

Virtual Labs

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Designing Test Suites

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Exercise 1

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Design a test suite for the following problem

The Absolute Beginners Inc. seems to have been fascinated by your work. Recently they have entrusted you with a task of writing a web-based mathematical software (using JavaScript). As part of this software, your team mate has written a small module, which computes area of simple geometric shapes. A portion of the module is shown below.

```
function square(side) { return side * side }

function rectangle(side1, side2) { return side1 * side2; }

function circle(radius) { return Math.PI * radius * radius; }

function right_triangle(base, height) { return 1 / 2 * base * height; }
```

Prepare a test suite that will

- Verify each of the above mentioned individual function is working correctly

Your task essentially is to verify whether each of the above function is returning correct values for given inputs. For example, a rectangle with length 10 unit and breadth 5 unit will have an area of 50 sq. unit. This can be verified from the output of the function call `rectangle(10, 5)`; However, testing also attempts to point out possible bugs in the software. How would the above code behave for a call `rectangle(10, -5)`;

Modify the code to address this defect.

- In each function, return -1 if any given dimension is negative
- Modify the test suite such that it reflects desired performance for both correct and incorrect input(s)

se-itkgp.vlabs.ac.in/exp/designing-test-suites/test_suit.html

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Code

function square(x) { return x*x }

+ Create Test Suite

Title

Summary

+ Add × Cancel

TS id	Title	Summary	Remove
TS0	Addition	Add two nos.	⊖
TS1	Calculate Square	Verify Square	⊖

⊕ Add Test Cases

Test Suite No.	Summary	Script	Expected Output	Actual Output	Manual Testing	Status
TS0	Add two nos.	sum(12,34)	46	46	☐ Yes	Pass

Test Suite id	Summary	Script	Expected Output	Actual Output	Manual Testing	Status
TS0	Test	sum(12,34)	46	46	☐ Yes	Pass
TS1	Verify square	square(10)	100	100	☐ Yes	Pass
TS2	Verify rectangle	rectangle(15,5)	75	78	☒ Yes	Fail
TS3	Verify circle	circle(7)	154	154	☐ Yes	Pass
TS4	Right triangle	right_triangle(13,7)	45.5	45.5	☐ Yes	Pass
TS5	Circle(negative)	circle(-14)	-1	-1	☐ Yes	Pass

# of test cases passed	4
# of test cases failed	1
Total # of test cases	5
Test suite status	Failed

Test Suite id	Summary	Script	Expected Output	Actual Output	Manual Testing	Status
TS0	Test	sum(12,34)	46	46	☐ Yes	Pass
TS1	Verify square	square(10)	100	100	☐ Yes	Pass
TS2	Verify rectangle	rectangle(15,5)	75	75	☐ Yes	Pass
TS3	Verify circle	circle(7)	154	154	☐ Yes	Pass
TS4	Right triangle	right_triangle(13,7)	45.5	45.5	☐ Yes	Pass
TS5	Circle(negative)	circle(-14)	-1	-1	☐ Yes	Pass

# of test cases passed	5
# of test cases failed	0
Total # of test cases	5
Test suite status	Passed